

B Summary of Parameters of ReSolveD- α

We provide a unified view of the selected parameters of ReSolveD- α in the following Table 24. In particular, for each parameter set of ReSolveD- α -{160, 256, 384, 512}{s, f}, we re-iterate its RSD parameters from Table 3, the corresponding VOLEitH parameters from Table 4, and their target security levels backed by the estimator results in Table 18.

For clarity, we note that the parameters for ReSolveD- α are our *recommended* parameters, and the ones of ReSolveD- α' shown in Table 13 are *not* recommended. The parameters of ReSolveD- α' only serve to illustrate what is the performance gain if we do not consider the approximate birthday paradox attack [CCJ23] based on a linear-time approximate collision search algorithm that is currently unknown.

Table 24: Summary of all parameter sets of ReSolveD- α . The RSD parameters m , k , w , and ℓ_{bs} denote the code length, code dimension, noise weight, and block size, respectively. The VOLEitH parameters ℓ_{vole} , τ , w_{grind} , and T_{open} denote the length of a VOLE commitment, number of VOLE instances, level for proof-of-work, and threshold for the BAVC opening size, respectively. The target quantum and classical security levels required by the Next-generation Commercial Cryptographic Algorithms Program (NGCC) [Ins25] are listed, and the concrete security of RSD instances are reported in Table 1 and Table 18.

Parameter set	RSD Parameters				VOLEitH Parameters				NGCC Security Level	
	m	k	w	ℓ_{bs}	ℓ_{vole}	τ	w_{grind}	T_{open}	Quantum	Classical
ReSolveD- α -160s	1860	1056	310	6	1056	14	6	129	80	128
ReSolveD- α -160f	1860	1056	310	6	1056	21	8	139	80	128
ReSolveD- α -256s	2988	1698	498	6	1687	22	12	224	128	256
ReSolveD- α -256f	2988	1698	498	6	1687	35	8	223	128	256
ReSolveD- α -384s	4452	2532	742	6	2510	34	10	332	192	384
ReSolveD- α -384f	4452	2532	742	6	2510	53	9	336	192	384
ReSolveD- α -512s	5940	3378	990	6	3343	46	6	439	256	512
ReSolveD- α -512f	5940	3378	990	6	3343	72	8	447	256	512